

Digital Research Toolkit for Linguists

Week 9: Creating reports with Quarto and knitr

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Psycholinguistics and Cognitive Modeling Lab

Homework

Questions?

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Documentation

When you trying to look at
the code you wrote a month ago

IT'S SOME KIND OF ELVISH

I CAN'T READ IT

imgflip.com

Why bother?

Someone WILL revisit your work → you or someone else

Up to date	keep track of all aspects
Reconsider	you think about what you're doing
Improve	quality (data, code, analysis, etc.)
Share	communication and collaboration
Remember	decisions, procedures, options, details, etc.
Reproducibility and transparency	easy access to information
Troubleshooting	what isn't doing its job



Jakob 🍀 \u0000

@jcsrb

6 hours of debugging can save
you 5 minutes of reading
documentation

ProgrammerHumor.io

Make your documentation

Understandable	to your audience (at least yourself)
Short	tl;dr
Descriptive	what is x and what does it do
Focused	get to the point
Complete	what data, packages, etc.
Fresh	memory decays fast
Timestamped	when did this happen?

How to document

BILLY

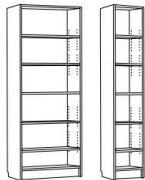
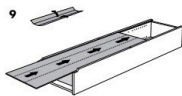
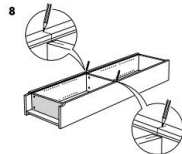
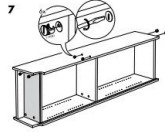
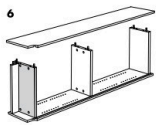
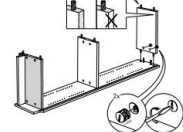
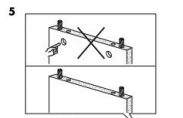
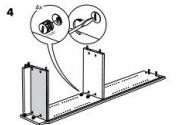
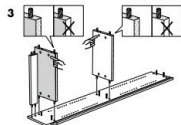
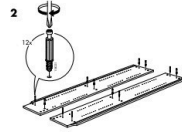
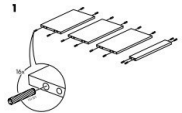
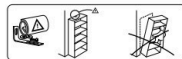
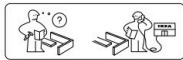


Image and text
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- You're not even making any sense.

Exact Instructions Challenge - THIS is why my kids hate me. | Josh Darnit

Josh Darnit 121K subscribers

43K Share Clip Save

Markdown, Pandoc, and Quarto

Markdown



is a lightweight markup language for creating easy to read and easy to write plain text files. Widely used (Wikipedia, ILIAS, blogs, forums, instant messaging, GitHub, software documentation etc.).

Key goal is **readability** (compare Word, RTF or HTML).

Reporting and Documenting with Quarto

Introduction

This a Quarto document. It shows a lot of features. You can use the source or visual editors to write your report and the render the file to the format of your choosing. Quarto primarily works with UTF-8 encoded text.

Structure

Give your file structure. Use sections to organize the content into logical bits.

Basic text formatting

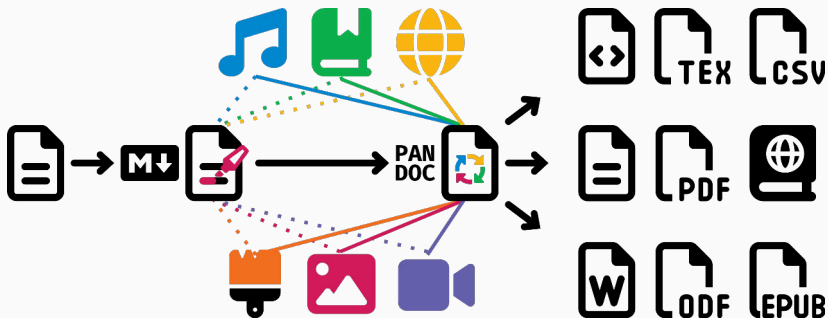
You can make your text stand out by making it ***bold***. Spacing matters: `\* \*this is broken\*`. In the visual editor, a lot of the options are available as the usual keyboard shortcuts.

Pandoc

Enhanced Markdown (e.g. supports bibliographies) with basic functionality (looks are less important).

It's a free and open source document/markup format converter: it converts a document from one format to another:

<https://pandoc.org/diagram.svgz?v=20240511135422>



 **quarto** is an open-source tool for creating dynamic documents, presentations, and websites.

It supports many programming languages/formats and can integrate **code**, *formatted* TEXT, and visualizations.

Pandoc markdown with extra syntax and features (e.g. code, interactivity). You write the document as QMD and export it as HTML, PDF, DOCX, ...

```
'''{r Participant duration plot}
# label: fig-part
# fig-cap: Experiment 2 duration
qqplot(part.summary) +
  aes(x = DURATION) +
  geom_histogram(bins = 30L, fill = bg_color,
  colour=bq_color, aes(v = after_stat(density))) +
  geom_density(adjust = 1L, fill = mn_color, colour =
  mn_color, alpha = 0.5) +
  labs(x = "Duration in minutes", y = "Density", title =
  "Distribution of experiment durations", subtitle="From
  the earliest time stamp during the ethics statement to
  the end of the experiment")
'''

'r nrow(part.summary)' participants took part in the
experiment (@tbl-part).
'r nrow(filter(part.summary, GENDER == "weiblich"))'
participants were women, and 'r
nrow(filter(part.summary, GENDER == "divers"))' was
non-binary.
The mean age was 'r round(mean(part.summary$AGE), 1)'
('r min(part.summary$AGE) -- r
max(part.summary$AGE)').
They took 'r round(mean(part.summary$DURATION),1)' on
average to complete the experiment (@fig-part). Some
participants were unrealistically fast at this task.
Those who took under 20 minutes are marked in orange. Of
all participants, 'r nrow(filter(part.summary, GERMAN !=
"ja"))' admitted to not being a native speaker of
```

A Concurrent Comparison of EyeLink 1000+ and TrackPixx3 on a Benchmark Test Battery

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Semantics vs agreement: What determines the interpretation of German bracketing paradoxes?

AMLaP 2024, September 5th 2024

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Examples

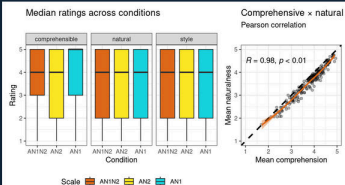
Contents

- Background
- Participants
- Analysis
- Responses
- Scales
- Items
- Morphosyntactic Annotation
- Inferential Statistics
- Response Time
- Metadata

CONDITION	MEAN	SD	SEM
AN1N2	3.61	0.36	0.06
AN2	3.54	0.28	0.05
AN1	3.60	0.29	0.05

Scales

Naturalness, comprehensibility, and grammaticality are often used interchangeably. Stylistic form could be related to the two. As it turns out, the scales were highly correlated. Overall, the ratings on the comprehensibility scale were higher than those for naturalness, which were higher than those for stylistic form (see [Table 4](#) and [Figure 2](#)).



More examples: <https://quarto.org/docs/gallery/>

Questions?

Wrap-up

- ✓ Documentation and why it's important
- ✓ Markdown and Pandoc
- ✓ Quarto intro
- ▶▶ **no class next week!**
- ▶▶ Quarto outro, text editors, writing reports
- ❓ Complete assignment 9 (→ ILIAS)